



مدارس
للتكنولوجيا
التطبيقية

تطوير المواقع والبرمجيات الصف الثالث

IT Department

Web Development and Programming

Web Development and Programming
Question Bank

2st

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تطوير المواقع والبرمجيات الصف الثالث

Unit	24
Name	Introduction to Software Testing
Goals / Outcomes	By the end of this unit, students should be able to: ➤ <u>Remembering</u> 1. Recall the definition of software testing, its phases, levels, and key types (e.g., manual vs. automated, API testing, functional vs. non-functional testing). 2. Identify key principles and tools for software testing (e.g., defect clustering, Agile Testing Quadrants, API testing tools like Postman). ➤ <u>Understanding</u> 1. Explain why software testing is critical in software development and distinguish between different testing methods (manual vs. automated, functional vs. non-functional). 2. Understand the Software Development Life Cycle (SDLC), Software Testing Life Cycle (STLC), and the role of Agile in testing. 3. Describe the significance of different testing levels (unit, integration, system, acceptance) and bug tracking processes. ➤ <u>Applying</u> 1. Apply software testing principles and stages in various testing scenarios. 2. Implement STLC and SDLC models, using tools like Postman for API testing or bug tracking systems to manage defects. 3. Use Agile testing practices during development sprints. ➤ <u>Analyzing</u> 1. Compare different testing approaches (manual vs. automated, API vs. GUI, Agile vs. traditional). 2. Differentiate between testing levels and methods (black-box, white-box, gray-box) and their impacts on software quality. ➤ <u>Evaluating</u> 1. Evaluate the effectiveness of early testing, different testing methods, and API testing in identifying defects. 2. Assess the impact of Agile KPIs (e.g., velocity, defect rate) and test case effectiveness. ➤ <u>Creating</u>

	1. Design comprehensive test plans and test cases for different software levels and features, including API testing. 2. Develop test scenarios and automated testing scripts, incorporating SDLC, STLC, or Agile methodologies.	
Knowledge	Code	Knowledge
	TPK11	software development
	TPK12	software testing approaches (analytical, model based, methodical...etc)
	TPK13	The basics of agile testing And API testing
Skill	Code	Skill
	TPC3.1	Use a APIs
	TPC3.2	Develop an application
	TPC3.3	Secure an application
	TPC3.4	Discover bugs in a software
	TPC3.5	Classify design patterns
	TPC3.6	Implement design patterns concepts into the projects
	TPC5.5	Creating an authentication system for your application
	TPC5.8	Creating and using composer packages

المعرفة	الكود
software development	TPK11

Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement.

No	Question	Answer
	Software testing ensures that a product is free of bugs and performs as required.	
	Testing is only about running the software and looking for bugs.	
	Verification checks if the software meets users' needs.	
	Debugging is the process of finding bugs, while testing is the process of fixing them.	
	Quality Assurance (QA) focuses on preventing bugs before they occur.	
	A defect in the code always leads to a failure when the software runs.	
	Exhaustive testing means testing every possible input combination.	
	Early testing helps reduce costs by catching bugs before they escalate.	
	The pesticide paradox suggests that repeating the same tests always finds new bugs.	
	Testing is the same for all types of software (e.g., games vs. banking apps).	
	Regression testing ensures that new changes don't break existing functionality.	

	Unit testing checks how different software components work together.	
	Performance testing checks how fast and stable software is under different workloads.	
	Static testing involves running the software to find defects.	
	Exploratory testing follows a strict script with predefined steps.	
	DevOps combines development and operations to deliver software faster.	
	Shift-Left Testing means delaying testing until the end of development.	
	A retrospective meeting focuses only on what went wrong in a project.	
	A test case includes steps, expected results, and actual results.	
	test scenario is more detailed than a test case.	
	A syntax error occurs when the code logic is incorrect but runs without crashing.	
	Jira is a popular bug-tracking tool used in Agile projects.	
	The Waterfall model allows revisiting previous phases after completion.	
	The Agile model focuses on early and continuous delivery of software.	
	User Acceptance Testing (UAT) is performed by developers.	

Dear learner: Choose the correct answer.

No	Question	Answer
	What is the main goal of software testing? a) To write code b) To ensure software works as expected and is bug-free c) To design user interfaces d) To manage project budgets	
	Which of the following is NOT a benefit of software testing? a) Early bug detection b) Increased development costs c) Improved software quality d) Meeting user expectations	
	Verification in software testing checks: a) If the software meets user needs b) Only the GUI design c) If the software meets specified requirements d) Only performance issues	
	What is the difference between testing and debugging? a) Testing fixes bugs; debugging finds them b) Testing finds bugs; debugging fixes them c) Both are the same d) Testing is done after debugging	
	Which principle states that most bugs are found in a few specific modules? a) Defect clustering b) Early testing c) Pesticide paradox d) Exhaustive testing	

	Which testing type checks individual units or components of code? a) Integration testing b) System testing c) Unit testing d) Acceptance testing	
	What does the "pesticide paradox" imply in testing? a) Repeating the same tests finds new bugs b) New tests are needed to find new bugs c) Bugs are like pests that never go away d) Automated tests are better than manual tests	
	Which phase of STLC involves creating test cases? a) Requirement Analysis b) Test Planning c) Test Case Design d) Test Execution	
	Which of the following is a non-functional test? a) Unit testing b) Performance testing c) Integration testing d) Regression testing	
	What is the purpose of regression testing? a) To test new features only b) To replace unit testing c) To test only the user interface d) To ensure new changes don't break existing functionality	
	Which tool is used for automated browser testing? a) Selenium b) Jira	

	c) Bugzilla d) Postman	
	What does Shift-Left Testing emphasize? a) Delaying testing until deployment b) Testing early in the development cycle c) Only manual testing d) Skipping unit tests	
	Which SDLC model delivers software in increments? a) Waterfall b) V-Model c) Incremental d) Spiral	
	What is the key focus of Agile methodology? a) Rigid planning b) Early and continuous delivery c) No customer involvement d) Avoiding testing	
	Which testing is done without predefined test cases? a) Regression testing b) Exploratory testing c) Unit testing d) System testing	
	What is the main purpose of a retrospective meeting? a) To assign blame for bugs b) To plan the next project c) To reflect on improvements d) To skip testing	
	Which component is NOT part of a test case? a) Test Case ID b) Expected Result	

	c) Developer's salary d) Test Steps	
	A bug that causes incorrect results without crashing is called: a) Syntax error b) Runtime error c) Logic error d) Security bug	
	Which tool is used for tracking bugs? a) Selenium b) Jira c) Postman d) Jenkins	
	Which testing phase involves end-users? a) Unit testing b) Integration testing c) UAT (User Acceptance Testing) d) Regression testing	
	What does CI/CD stand for in DevOps? a) Continuous Inspection/Continuous Deployment b) Continuous Integration/Continuous Delivery c) Code Inspection/Code Deployment d) Customer Input/Customer Delivery	
	Which SDLC model is sequential and non-iterative? a) Agile b) Waterfall c) Spiral d) Iterative	
	What type of bug affects software security? a) Usability bug	

	b) Performance bug c) Security bug d) Syntax bug	
	Which testing checks software compatibility across devices? a) Performance testing b) Compatibility testing c) Regression testing d) Unit testing	
	Who performs unit testing? a) Testers b) End-users c) Developers d) Business analysts	

Dear learner: Give The Scientific Term Of These Sentence

No	Question	Answer
	The process of ensuring software is free of bugs and works as expected is called _____.	
	Checking if the software meets specified requirements is known as _____.	
	The phase in STLC where test cases are created is called _____.	
	A set of steps to verify a feature's functionality is called a _____.	
	Testing individual units of code is known as _____.	
	The principle that states most bugs are found in a few modules is called _____.	
	Repeating the same tests and finding fewer bugs	

	over time is called the _____.	
	The model that delivers software in small, usable increments is the _____ model.	
	A meeting to discuss project improvements is called a _____.	
	A mistake in code that causes incorrect behavior is called a _____.	
	The tool used to track and manage bugs is called a _____.	
	Testing software without predefined test cases is called _____ testing.	

Match each SDLC model with its description:

No	Model	No.	Description	Answer
1	Waterfall	A	Delivers parts of the software quickly for early user feedback.	
2	V-Shaped	B	Combines iterative and incremental approaches, focusing on adaptability.	
3	Iterative	C	Follows strict sequential phases with heavy documentation.	
4	Incremental	D	Repeats cycles of development and testing to refine the product.	
5	Agile	E	Emphasizes testing at every development phase.	
6	Spiral	F	Focuses on risk analysis and iterative refinement (not mentioned but plausible for advanced testing).	

Match each testing type with its purpose:

No	Testing Type	No.	Purpose	Answer
1	Unit Testing	A	Ensures the entire system works as expected.	
2	Integration Testing	B	Checks how different modules interact.	
3	System Testing	C	Tests individual functions or components.	
4	UAT (Acceptance)	D	Evaluates security vulnerabilities.	
5	Regression Testing	E	Confirms software meets user requirements before release.	
6	Security Testing	F	Re-runs old tests to ensure new changes don't break existing features.	

Match each term with its correct definition:

No	Term	No	Definition	Answer
1	Testing	A	Fixing identified defects in the code.	
2	Debugging	B	Detecting defects in the software.	
3	Verification	C	Ensuring the software meets user needs.	
4	Validation	D	Checking if the software meets specified requirements.	
5	Error	E	A mistake made by a developer.	
6	Defect	F	A flaw in the software causing incorrect behavior.	

Match each test case element with its description:

No	Component	No	Description	Answer
1	Test Case ID	A	Detailed steps to execute the test.	
2	Description	B	What should happen if the feature works correctly.	
3	Preconditions	C	Unique identifier for the test case.	
4	Test Steps	D	Input values used in the test.	
5	Expected Result	E	Requirements before the test can run.	
6	Test Data	F	What feature or function is being tested?	

Dear Learner: Answer the following questions.

	Question	What is software testing?
	Question	What is the difference between verification and validation?
	Question	What is the difference between testing and debugging?
	Question	What is the pesticide paradox in testing?
	Question	Name two test activities in the STLC.
	Question	What is the main goal of a retrospective in Agile?

	Question	What is a test case?
	Question	Name two common types of bugs.
	Question	What is Agile testing?

المعرفة	الكود
software testing approaches (analytical, model based, methodical...etc)	TPK12

Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement.

No	Question	Answer
	Unit testing checks how different modules work together.	
	System Integration Testing (SIT) ensures seamless communication between different systems.	
	Alpha testing is performed by real users in a production environment.	
	Black-box testing requires knowledge of the internal code structure.	
	Boundary Value Analysis (BVA) tests values at the edges of input ranges.	
	Decision table testing is used when there are multiple conditions and actions.	
	State transition testing is used when the system behavior changes based on different states.	
	White-box testing focuses only on the software's external behavior.	
	Statement coverage ensures all branches in the code are executed.	
	Error guessing relies on predefined test cases rather than tester experience.	
	Exploratory testing follows a strict script.	

	Checklist-based testing is best suited for regression testing.	
	User stories should follow the INVEST rule, where "T" stands for "Timely."	
	Acceptance criteria define when a user story is considered complete.	
	Collaborative user story writing involves only developers, not testers.	

Dear learner: Choose the correct answer.

No	Question	Answer
	Which testing level focuses on individual code components? a) System Testing b) Unit Testing c) Acceptance Testing d) Integration Testing	
	What is the main purpose of System Integration Testing (SIT)? a) Test individual modules b) Verify interactions between different systems c) Check GUI functionality d) Only validate database queries	
	Which testing technique divides input data into equivalent partitions? a) Boundary Value Analysis b) Decision Table Testing c) Equivalence Partitioning d) State Transition Testing	

	Which of the following is a Black-box testing technique? a) Statement Coverage b) Branch Coverage c) Boundary Value Analysis d) Error Guessing	
	What does BVA (Boundary Value Analysis) primarily test? a) Middle-range values b) Edge values of input ranges c) Only invalid inputs d) Database connections	
	Which technique uses tables to represent input combinations and outputs? a) State Transition Testing b) Checklist-Based Testing c) Exploratory Testing d) Decision Table Testing	
	What is the goal of White-box testing? a) Test internal code and logic b) Verify only GUI elements c) Ignore error handling d) Only validate user stories	
	Which coverage ensures all decision points in code are tested? a) Statement Coverage b) Branch Coverage c) Function Coverage d) Path Coverage	
	Which experience-based technique relies on tester intuition?	

	a) Error Guessing b) Decision Table Testing c) Equivalence Partitioning d) Boundary Value Analysis	
	What is the main focus of Exploratory Testing? a) Following strict test scripts b) Unscripted, real-time testing c) Only validating acceptance criteria d) Ignoring edge cases	
	Which part of a user story describes "why" the feature matters? a) Role b) Goal c) Benefit d) Acceptance Criteria	
	What does the "V" in INVEST stand for? a) Valuable b) Verified c) Variable d) Visible	
	Which of the following is NOT part of acceptance criteria format? a) Given b) When c) Then d) Why	
	Which testing approach improves team communication? a) Black-box Testing b) White-box Testing	

	c) Collaborative User Story Writing d) Error Guessing	
	What is the final testing phase before software release? a) Unit Testing b) System Testing c) Acceptance Testing d) Integration Testing	

Dear learner: Complete the question.

No	Question	Answer
	_____ testing checks individual code components in isolation.	
	_____ testing ensures different systems work together.	
	_____ testing is done by real users before the final release.	
	_____ divides input data into valid and invalid groups.	
	_____ ensures all decision branches in code are tested.	
	_____ testing relies on tester experience to guess bugs.	
	A _____ describes a feature from a user's perspective.	
	_____ criteria define when a user story is complete.	
	The _____ rule ensures user stories are well-defined.	

	_____ testing explores the app without predefined scripts.	
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Dear Learner: Match the testing level to its description.

No	Testing Level	No	Description	Answer
1	Unit Testing	A	Tests interactions between systems	
2	Integration Testing	B	Checks individual code components	
3	System Testing	C	Validates the full system	
4	System Integration Testing	D	Combines modules	
5	Acceptance Testing	E	Final approval by users	
6	Alpha Testing	F	Internal testing at developer's site	

Match the testing technique to its use case.

No	Technique	No	Use Case	Answer
1	Equivalence Partitioning	A	Tests edge values of inputs	
2	BVA	B	Covers all code statements	
3	Decision Table	C	Groups similar inputs	
4	State Transition	D	Tests multiple conditions/actions	
5	Statement Coverage	E	Checks system behavior across states	
6	Error Guessing	F	Uses tester intuition	

Dear Learner: Answer the following questions.

Question	What is the purpose of System Integration Testing (SIT)?
Question	What does "100% branch coverage" mean?
Question	Why are acceptance criteria important?
Question	What is the INVEST rule in user stories?
Question	What is checked in White-box testing?
Question	Name one advantage of checklist-based testing.

المعرفة	الكود
The basics of agile testing And API testing	TPK13

Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement.

No	Question	Answer
	Agile methodology follows a linear (waterfall) approach to development.	
	Sprints in Agile typically last 1-2 weeks.	
	The Agile Manifesto prioritizes processes and tools over individuals and interactions.	
	Agile testing starts only after development is complete.	
	Daily Scrums are optional in Agile teams.	
	Kanban uses fixed sprints like Scrum.	
	Extreme Programming (XP) emphasizes Test-Driven Development (TDD).	
	The Product Owner in Scrum sets task priorities.	
	Agile teams should avoid changing requirements once development starts.	
	Retrospectives help teams improve their processes.	
	API testing only checks the graphical user interface (GUI).	
	API testing helps detect bugs early in development.	
	GUI testing is faster than API testing.	
	API testing can only be done in Java.	

	Security testing is not part of API testing.	
	Load testing simulates a single user accessing an API.	
	Postman is a command-line API testing tool.	
	Unit testing checks how different APIs work together.	
	JSON and XML are common data formats in API testing.	
	API testing reduces costs by catching errors early.	

Dear learner: Choose the correct answer.

No	Question	Answer
	What is the primary focus of Agile methodology? a) Strict documentation b) Long development cycles c) Avoiding all changes d) Customer satisfaction and flexibility	
	Which Agile framework uses fixed-length sprints? a) Scrum b) Kanban c) Extreme Programming (XP) d) Waterfall	
	What is the main purpose of a Sprint Retrospective? a) To demo new features b) To improve team processes c) To assign new tasks d) To write documentation	
	Which role removes obstacles in Scrum? a) Product Owner b) Scrum Master	

	c) Developer d) Stakeholder	
	What does Test-Driven Development (TDD) involve? a) Writing tests after coding b) Skipping unit tests c) Writing tests before coding d) Only manual testing	
	Kanban uses: a) A visual board (To Do, In Progress, Done) b) Fixed sprints c) Only written reports d) No task tracking	
	What is the recommended duration for a Daily Scrum? a) 1 hour b) 15 minutes c) 30 minutes d) No time limit	
	Which Agile framework focuses on technical excellence? a) Scrum b) Kanban c) Extreme Programming (XP) d) Lean	
	What does API testing validate? a) Only UI design b) API behavior, performance, and security c) Database size d) User experience surveys	

	Which is NOT a type of API testing? a) Unit testing b) GUI testing c) Load testing d) Security testing	
	What does load testing simulate? a) A single API request b) Only manual testing c) Multiple users accessing an API d) Database queries	
	Which data formats are used in API testing? a) HTML and CSS b) XML and JSON c) PDF and DOCX d) MP3 and MP4	
	What is the main benefit of early bug detection in API testing? a) Saves time and money b) Increases costs c) Delays releases d) Requires more documentation	
	Which tool is a command-line API testing utility? a) Postman b) SoapUI c) curl d) JMeter	
	What does integration testing check? a) Single API components b) How different APIs work together c) Only security flaws d) UI responsiveness	

	Which testing ensures an API meets its intended purpose? a) Performance testing b) Load testing c) Validation testing d) Security testing	
	Why is API testing language-independent? a) It only tests UI b) It uses XML/JSON for data exchange c) It requires specific languages d) It ignores data formats	

Dear learner: Give The Scientific Term Of These Sentence (Fill-in-the-Blank Style)

No	Question	Answer
	_____ methodology breaks projects into short cycles called sprints.	
	_____ is a visual workflow management method used in Agile.	
	The _____ Manifesto prioritizes individuals and interactions over processes and tools.	
	_____ testing ensures APIs work correctly under heavy user traffic.	
	_____ is a popular GUI tool for API testing.	
	_____ is a command-line tool for sending HTTP requests.	
	_____ is the Agile role responsible for removing obstacles.	
	_____ testing checks individual API components.	
	_____ is an Agile framework emphasizing	

	technical excellence.	
	_____ is a data format commonly used in API testing.	
	_____ testing protects APIs from external threats.	
	_____ is a meeting where Agile teams review progress daily.	
	**Agile development uses short cycles called _____.	
	The _____ owns the product backlog in Scrum.	
	_____ testing involves writing tests before writing the actual code.	
	_____ is a tool used for performance and API testing.	
	API testing commonly uses _____ and _____ for data exchange.	
	The _____ ensures the team follows Scrum practices.	
	_____ testing checks how different APIs interact.	
	_____ is a lightweight alternative to Scrum with continuous workflow.	
	_____ is a security-focused type of API testing.	
	The _____ meeting helps Agile teams reflect on improvements.	

Match the Agile Framework to its Description

No	Framework	No.	Description	Answer
1	Scrum	A	Uses a visual board (To Do, In Progress, Done) with no fixed sprints.	
2	Kanban	B	Focuses on technical excellence with Pair Programming and Test-Driven Development (TDD).	
3	Extreme Programming (XP)	C	Breaks work into fixed-length sprints (2-4 weeks) with defined roles (PO, Scrum Master).	
4	None of the above	D	A hybrid approach combining Scrum and Kanban.	

Match the Agile Testing Life Cycle Phase to its Description

No	Phase	No.	Description	Answer
1	Impact Assessment	A	Daily meetings to update testing status and set priorities.	
2	Test Planning	B	Final checks to decide if features are ready for release.	
3	Daily Scrums	C	Gathering stakeholder input for future implementation.	
4	Test Agility Review	D	Preparing a testing schedule and expected outcomes.	
5	Release Readiness	E	Evaluating progress to ensure alignment with the plan.	

6	None of the above	F	Conducting security audits before deployment.	
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Match the API Testing Type to its Purpose

No	Testing Type	No.	Purpose	Answer
1	Unit Testing	A	Simulates multiple users accessing the API simultaneously.	
2	Integration Testing	B	Tests individual API components in isolation.	
3	Performance Testing	C	Checks how different APIs interact with each other.	
4	Load Testing	D	Evaluates speed, efficiency, and scalability.	
5	Security Testing	E	Protects against external threats and data breaches.	
6	Validation Testing	F	Verifies if the API meets its intended specifications.	

Dear Learner: Answer the following questions.

Question	What is the main goal of Agile methodology?
Question	Name two Agile frameworks.
Question	What is the purpose of a Daily Scrum?
Question	What does a Product Owner do in Scrum?
Question	Name two common data formats used in API testing.
Question	What tool is used for testing REST APIs?
Question	What is Test-Driven Development (TDD)?
Question	What is the purpose of a Retrospective in Agile?